

9	(a)	1) The <u>existence of a threshold frequency</u> below which no photoelectrons are emitted proves that <u>electromagnetic radiation (EM) consists of discrete quanta of energy given by hf</u>. [B1] 2) The <u>instantaneous emission</u> of photoelectrons when <u>all the photon energy is transferred immediately to the electron in a single collision</u> gives evidence to particulate nature of EM. [B1] 3) The <u>maximum kinetic energy of the photoelectrons</u> being <u>dependent only on frequency of radiation f, which relates to the discrete energy of photon, and independent on the intensity</u> of radiation also give evidence for the particulate nature of EM. [B1]	
	(b)	$eV_s = \frac{1}{2}mv^2$ $(1.6 \times 10^{-19})V_s = \frac{1}{2}(9.11 \times 10^{-31})(1.42 \times 10^6)^2 \quad [C1]$ $V_s = 5.74 \text{ V} \quad [A1]$	
	(c)	reverse the polarity of the voltage source (and p.d. to be at least 5.74 V) [B1]	
	(d)	(i)	$h \frac{c}{\lambda} = \phi + \frac{1}{2}mv^2$ $h \frac{c}{\lambda} = (2.80 + 5.74) \text{ eV}$ $6.63 \times 10^{-34} \frac{3.00 \times 10^8}{\lambda} = 8.54 \times 1.6 \times 10^{-19} \quad [C1]$ $\lambda = 1.46 \times 10^{-7} \text{ m} \quad [A1]$
		(ii)	Ultra-violet [B1] [since visible light is between $4.0 \times 10^{-7} \text{ m}$ to $7.0 \times 10^{-7} \text{ m}$]
	(e)	[B1]: The cathode is heated to <u>release electrons</u>. [B1]: These electrons are then <u>accelerated (speed up) via</u> the accelerating potential (p.d.) in the circuit to have high speeds. [B1]: when these high speed electrons strike the anode, electrons <u>experienced large deceleration</u>, thus producing X-ray photons. [B1]: the electrons decelerate to <u>different extents</u>, hence results in a range of X-ray photon energies produced [B1]: high speed electron stops abruptly in a single collision and <u>all the kinetic energy of the electron is completely converted to a single X-ray photon (with the smallest wavelength)</u>. that resulted in the continuous spectrum of X-ray spectrum.	

	(f)	Similarity [B1]: any of the following 1) presence of cathode and anode 2) vacuum is needed 3) potential difference (p.d.) is set up across 2 plates 4) electrons moving across cathode towards anode 5) a complete circuit Difference [B1]: any of the following 1) p.d. across photoelectric is only a few volts while p.d. across X-ray tube is hundred thousand volts. 2) speed of electrons emitted (from cathode) due to photoelectric effect is significantly lower than that of X-ray tube. 3) p.d. across photoelectric can be reverse (to find stopping potential) while p.d. across X-ray tube is one directional (for accelerate the electrons). 4) electrons emitted from cathode of photoelectric is due to one-to-one interaction of photon with electron, while electrons emitted from X-ray tube is through heating of cathode. 5) Source of reference: EM radiation is needed to demonstrate photoelectric effects while high speed electrons are needed to produce X-ray spectrum. 6) One absorbs photon, another produces photon.	
	(g)	(i)	$eV = h \frac{c}{\lambda_{\min}}$ $(1.6 \times 10^{-19})V = (6.63 \times 10^{-34}) \frac{3.00 \times 10^8}{16 \times 10^{-12}} \quad [\text{C1}]$ $V = 77.7 \times 10^3 \text{ V} \quad [\text{A1}]$

		(ii) Relative intensity of K_{α} line comprise of contribution from both the continuous spectrum and characteristic peaks. a) Difference in intensities between the two K_{α} lines is primarily due to a <u>difference in the continuous spectrum of tungsten and barium</u>. [B1] AND b) The continuous radiation contribution for tungsten is more because tungsten has a <u>larger and more positive nuclei</u> as compared to Barium. Hence <u>more electrons passing in the vicinity of the tungsten nuclei are likely to deflect and give out an x ray photon</u>. [B1] Alternatively, tungsten has a <u>larger and more positive nuclei</u>. Thus for an <u>electron at a fixed distance</u> from the nuclei, the electron is likely to <u>deflect more and give out a higher energy x ray photon</u>. The intensity <u>of the high energy x ray photons for tungsten is thus larger</u>. [B1] Thus the continuous radiation contribution for tungsten is more. OR: Higher Z, stronger attractive force. Hence, larger braking effect Higher Z, target area is increased. Hence, larger probability of electrons striking.—leading to higher intensity of X ray produced. Thus the continuous spectrum of tungsten has a higher intensity as compared to Barium.
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